

Adversarial Attacks

Robust Deep Learning (Seminar)

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1 Introduction

Neural Networks ...

- what are they
- general robustness problem/
- descr. of adv. examples, ...? !Combine with early adv. history
- relevance
- structure of this report

[BRIEF neural network + backpropagation introduction?]

In deep learning, networks with many layers are used to solve tasks, such as image classification, in which we call the network the classifier.

2 Adversarial Attacks

2.1 Definition

Adversarial Attacks can be described as inputting *adversarial examples* into a machine learning system. [brendel2017decision] These two terms were introduced by [szegedy2013intriguing] and have since been used in different variations, however we here use the following definition:

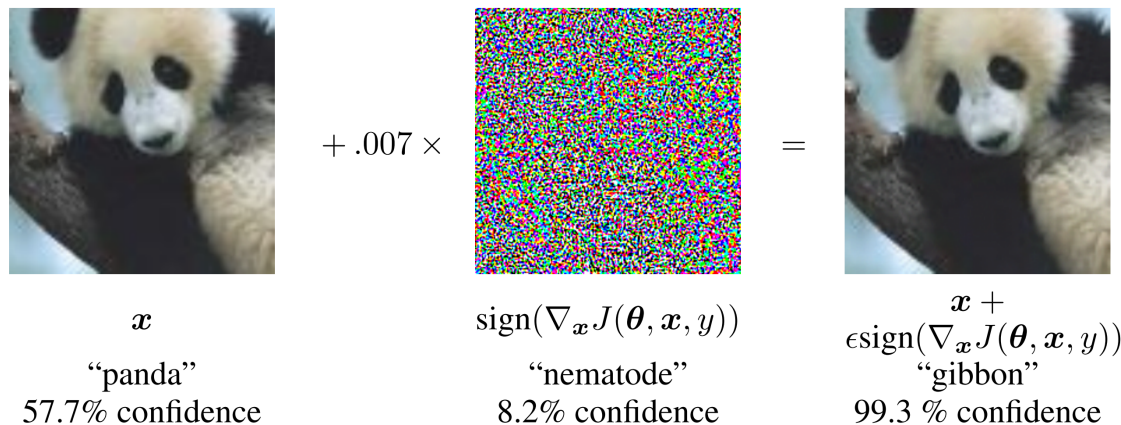


Figure 1: Classic illustration of an adversarial example. A small perturbation that is nearly imperceptible to humans is added to an image of a panda. While the original image is correctly classified as a panda with high confidence, the perturbed image is misclassified as a gibbon; also with high confidence. Adapted from [goodfellow2014explaining].

Definition 1 (Adversarial Example). An adversarial example is an input to a machine learning model that has been intentionally perturbed, typically in a subtle and often imperceptible manner, such that the model produces an incorrect prediction while the semantic content of the input remains unchanged.

Machine Learning Models. While adversarial examples are most commonly studied in the context of deep neural network classifiers, [szegedy2013intriguing; madry2017towards; pitropakis2019taxonomy] the phenomenon is not limited to neural networks. Adversarial vulnerabilities have been demonstrated across a wide range of machine learning architectures, including decision trees, support vector machines, k -nearest neighbour classifiers, logistic regression models, ensemble methods and more. This suggests that susceptibility to adversarial manipulation is a general property of learned decision functions rather than a characteristic of a specific machine learning method. [papernot2016distillation; szegedy2013intriguing; goodfellow2014explaining]

Intentional Perturbations. The requirement that perturbations are *intentional* reflects the modern understanding of adversarial examples. The seminal work of Szegedy et al. [szegedy2013intriguing] demonstrated the existence of inputs that caused misclassification despite appearing unchanged to humans, but did not initially frame these inputs primarily as the result of a deliberate attack. As the field evolved, researchers showed that such inputs can be generated systematically through optimization procedures and exploited by an adversary. [moosavi2016deepfool; madry2017towards] Consequently, adversarial examples are now typically discussed in the context of adversarial attacks, where perturbations are deliberately crafted to induce erroneous model behaviour. [chakraborty2018adversarial; pitropakis2019taxonomy; costa2024deep]

Subtle and Imperceptible Perturbations. The notion of *subtle* perturbations can be interpreted in two related but distinct ways. First, adversarial perturbations are commonly constrained to be mathematically small according to a chosen distance metric. The most frequently used measures are the L_0 norm, which counts the number of modified features or pixels; the L_2 norm, which measures the Euclidean distance between the original and perturbed inputs; and the L_∞ norm, which limits the maximum change applied to any individual feature. [costa2024deep] These norms define the perturbation budget available to an adversary and provide a mathematical notion of similarity between inputs. We will discuss these different norms in more detail in subsection 2.2.

Second, perturbations are often required to be imperceptible to humans. In the image domain, this means that the perturbed input should be visually indistinguishable or at least hardly distinguishable from the original image, preserving its semantic meaning. [madry2017towards; moosavi2016deepfool] Although we rarely see the perceptibility of the applied perturbations rigorously validated—for example, through experiments with human subjects or using LPIPS [zhang2018unreasonable] as a proxy metric—it is nevertheless very common for authors to state, based on a visual assessment, that the adversarial examples they have generated are indistinguishable from the original to humans. [brendel2017decision; chakraborty2018adversarial; costa2024deep; liang2022adversarial; papernot2016distillation; papernot2016transferability]

Confidence of Misclassification. Some definitions additionally require adversarial examples to induce misclassifications with high confidence. [madry2017towards] In practice, however, high-confidence errors are often a consequence of the optimization objective used during adversarial example generation rather than a defining characteristic of adversarial examples themselves. For example, early optimization-based attacks explicitly maximize the model loss or the probability assigned to a target class. [szegedy2013intriguing] Consequently, contemporary literature generally treats high confidence as an attack objective rather than a necessary component of the definition.

2.2 Taxonomy

Adversarial attacks can be categorised according to several orthogonal dimensions, including the attacker’s knowledge of the target model, the stage at which the attack is performed, the attacker’s objective, and the constraint that bounds the perturbation. These dimensions are widely used throughout the literature and provide a useful framework for comparing attack methodologies. [pitropakis2019taxonomy; costa2024deep] Figure 2 summarises these four dimensions together with their possible values; the attacks discussed in this survey are classified according to this taxonomy, with a complete overview provided in Table ??.

By Attacker Knowledge One of the most common distinctions is based on the amount of information available to the attacker about the target model. [pitropakis2019taxonomy; costa2024deep]

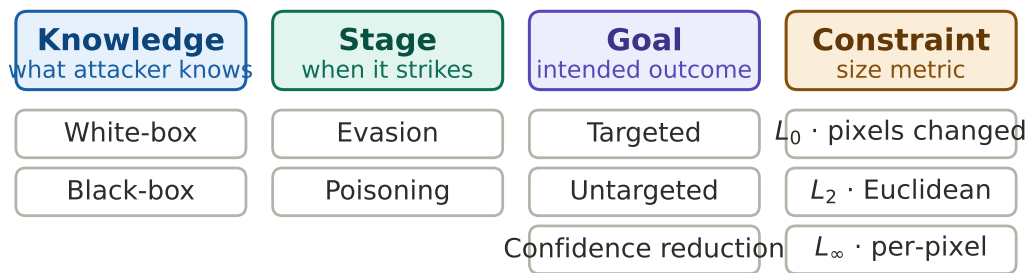


Figure 2: The four orthogonal dimensions along which adversarial attacks are classified: the attacker’s *knowledge* of the target model, the *stage* of the machine-learning pipeline at which the attack acts, the attacker’s *goal*, and the *constraint* (distance metric) that bounds the perturbation. A given attack is characterised by one choice from each column.

White-box Attacks. In a white-box setting, the attacker has complete knowledge of the target model, including its architecture, parameters, training procedure, and gradients. This level of access enables highly effective optimisation-based attacks that directly exploit gradient information to identify adversarial perturbations. Many influential attacks, including the Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and the Carlini–Wagner attack, were originally proposed in this setting. White-box attacks are commonly used to evaluate the worst-case robustness of machine learning models.

Black-box Attacks. In a black-box setting, the attacker has no direct access to the internal structure of the model and can only observe its outputs in response to supplied inputs. Such attacks more closely resemble many real-world scenarios, where proprietary models are exposed only through prediction interfaces or APIs. Black-box attacks typically rely on repeated queries, surrogate models, or the transferability of adversarial examples between different models.

By Attack Stage Attacks can also be classified according to the stage of the machine learning pipeline at which they are carried out.[chakraborty2018adversarial; pitropakis2019taxonomy]

Evasion Attacks. Evasion attacks are performed during the inference phase after a model has been trained. The attacker modifies input samples in order to induce misclassification while avoiding detection. Since they require no influence over the training process, evasion attacks are often considered the most practical and widely studied form of adversarial attack. Examples include adversarial images that cause image classifiers to produce incorrect predictions and manipulated network traffic designed to evade intrusion detection systems.

Poisoning Attacks. Poisoning attacks target the training phase by injecting maliciously crafted samples into the training dataset. The objective may be to degrade the model’s overall performance, manipulate its behaviour for specific inputs, or implant hidden backdoors that can later be triggered by an attacker. While poisoning attacks require some influence over the data collection or training pipeline, this assumption is realistic in settings that rely on crowdsourced, user-generated, or automatically collected data. Examples include recommendation systems trained on user feedback, spam filters trained on submitted emails, and machine learning models trained on large datasets scraped from the internet.

By Attack Goal A third dimension considers the attacker’s intended outcome.[chakraborty2018adversarial; costa2024deep; brendel2017decision]

105 *Targeted Attacks.* Targeted attacks seek to force the model to produce a specific attacker-chosen
 106 output. For example, an attacker may attempt to modify an image of a panda such that it is classified
 107 as a gibbon. Because the attack must drive the prediction toward a particular target class, targeted
 108 attacks are generally more difficult to perform than untargeted attacks.

109 *Untargeted Attacks.* Untargeted attacks aim only to cause a prediction error, but do not specify
 110 the resulting incorrect class. The attack is therefore successful as soon as the model's prediction
 111 differs from the correct label. The attacker might aim at having the classifier recognize the image as
 112 showing anything but a panda.

113 *Confidence Reduction Attacks.* Rather than inducing a specific misclassification, confidence reduction
 114 attacks aim to decrease the model's confidence in its predictions. Such attacks may not necessarily
 115 change the predicted class but instead increase uncertainty or reduce the probability assigned to the
 116 correct label. These attacks are particularly relevant in safety-critical applications, where confidence
 117 estimates are used to support downstream decision-making. By lowering confidence, an attacker
 118 may trigger fallback mechanisms, human intervention, or rejection of otherwise valid inputs.

119 **By Perturbation Constraint** Adversarial attacks are commonly categorised according to the distance
 120 metric used to constrain the perturbation applied to an input.[costa2024deep; liang2022adversarial]
 121 These constraints define what constitutes a *small* modification and therefore determine the attacker's
 122 search space.[carlini2017towards] The majority of attacks in the literature operate under one of
 123 three norm-based threat models. Figure 3 illustrates these three norms geometrically.

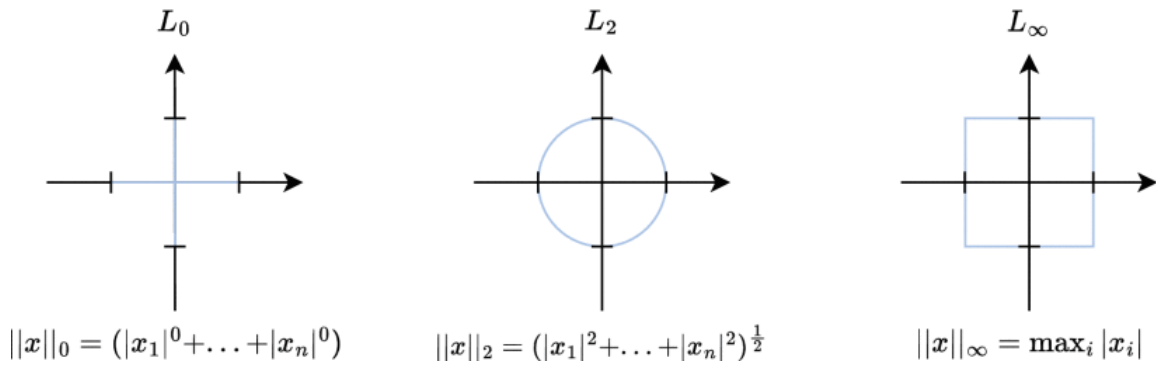


Figure 3: Geometric interpretation of the three norm constraints commonly used in adversarial machine learning. The unit balls of the L_0 , L_2 , and L_∞ norms are shown in a two-dimensional feature space. The L_0 norm constrains the number of modified features, the L_2 norm constrains the Euclidean distance between the original and perturbed input, and the L_∞ norm constrains the maximum change applied to any individual feature. Figure adapted from [costa2024deep].

124 *L_0 -Constrained Attacks.* The L_0 norm measures the number of features that have been modified. In the
 125 context of image classification, this corresponds to the number of altered pixels regardless of the mag-
 126 nitude of the changes. Attacks operating under an L_0 constraint aim to minimise the number of mod-
 127 ified features while still causing misclassification.[costa2024deep; liang2022adversarial] Examples
 128 for such attacks include Deepfool [moosavi2016deepfool] and Carlini & Wagner.[carlini2017towards]

129 *L_2 -Constrained Attacks.* The L_2 norm measures the Euclidean distance between the original and
 130 perturbed input vectors. Under this constraint, perturbations may be distributed across many
 131 features as long as the overall magnitude remains small.[costa2024deep; liang2022adversarial] The
 132 seminal work of Szegedy et al. [szegedy2013intriguing] formulated adversarial example generation
 133 as an optimisation problem under an L_2 constraint. Due to its intuitive geometric interpretation, the
 134 L_2 norm remains widely used when evaluating adversarial robustness.[costa2024deep]

135 *L_∞ -Constrained Attacks.* The L_∞ norm limits the maximum change that can be applied to any
 136 individual feature.[costa2024deep; liang2022adversarial] In image domains, this ensures that every
 137 pixel is modified by at most a small amount. Many influential attacks, including FGSM and PGD,
 138 operate under an L_∞ constraint because it provides a straightforward approximation of perceptual
 139 similarity while enabling efficient optimisation.[goodfellow2014explaining; madry2017towards]

140 **Why do adv. examples exist? (Draft!)** When adversarial examples were first empirically docu-
 141 mented in 2013 [cite], it was first hypothesized that they were caused by a supposedly highly
 142 non-linear nature of neural networks. In 2014, Goodfellow et al.[cite] argued that they are instead a
 143 result of overly linear behaviour.

144 They considered a simple linear regression model [formula] and L_∞ perturbations. [Formula:
 145 separate x , perturbation] In high dimensions, if we choose perturbation=..., [term] can get large
 146 enough to impact the model prediction. By linearizing the loss function[see section below], they
 147 were able to construct adversarial examples on neural networks as well. On the contrary, models
 148 like RBF did not seem as vulnerable to adversarial examples, further giving support to their theory.

149 More detailed papers like [Adversarial examples are not bugs, they are features] -> Is it even
 150 possible to have networks not vulnerable to adversarial examples?

151 2.3 Notable Evasion Attacks

152 This section presents several notable adversarial attacks that have significantly influenced the
 153 development of adversarial machine learning. An overview of these attacks, together with their
 154 classification according to the taxonomy introduced in Section ??, is provided in Table ??.

155 **L-BFGS** The first widely recognised adversarial attack was introduced by Szegedy et al. [szegedy2013intriguing].
 156 Their work did not only show that machine learning models are vulnerable to inputs that lead
 157 to incorrect predictions while appearing unchanged to human observers, first describing the phe-
 158 nomenon of an adversarial example. They further introduced a fairly reliable method to find such
 159 examples.

160 To generate adversarial examples, the authors formulated the problem as a constrained optimi-
 161 sation task. Given an input x and a target class l , the objective is to find the smallest perturbation r
 162 such that the perturbed input $x + r$ is classified as the target class:

$$163 \min_r \|r\|_2 \quad \text{subject to} \quad f(x + r) = l, \quad (1)$$

164 where f denotes the classifier and $\|r\|_2$ represents the Euclidean distance between the original
 165 and perturbed input. Directly solving this problem is difficult because the classification constraint
 166 $f(x + r) = l$ is highly non-linear and non-convex. To make the problem tractable, Szegedy et al.
 167 reformulated it as a continuous optimisation problem by replacing the hard classification constraint
 168 with a differentiable loss term that encourages the model to predict the target class. Specifically, they
 169 minimise

$$170 \min_r c\|r\|_2 + \text{loss}_f(x + r, l), \quad (2)$$

171 subject to the box constraint

$$172 x + r \in [0, 1]^m, \quad (3)$$

173 where $\text{loss}_f(x + r, l)$ measures how strongly the network predicts class l , $c > 0$ controls the
 174 trade-off between achieving the target misclassification and keeping the perturbation small, and the
 175 box constraint ensures that the perturbed input remains a valid image. This reformulation yields
 176 a differentiable objective whose gradients can be computed via backpropagation, allowing the
 177 perturbation to be found efficiently using the box-constrained Limited-memory Broyden–Fletcher–
 178 Goldfarb–Shanno (L-BFGS) optimisation algorithm. A line search over c is then performed to identify
 179 the smallest successful perturbation that causes the classifier to output the target class.

Although the L-BFGS attack is still computationally expensive compared to later methods, its significance lies in establishing the existence of adversarial examples and motivating the extensive research that followed. For this reason, it is widely regarded as the starting point of adversarial machine learning research. [goodfellow2014explaining; costa2024deep; liang2022adversarial]

Fast Gradient Sign Method (FGSM) (Draft!) Goodfellow et al hypothesized that the reason adversarial examples exist is that commonly, models are designed to behave mostly linear [goodfellow2014explaining]. Previously, it was suspected that models were too non-linear [cite].

Based on this hypothesis, they came up with the fast gradient sign method to find L_∞ -constricted adversarial examples. L_∞ is chosen because it models sensor inaccuracies: a small perturbation in a sensor value would still be encoded to the same digital value. Unlike L-BGFS, they did not choose a specific target class they want the adversarial example to be mistaken for. Instead, they aim to maximize the loss value of the adversarial example. That means that we want to solve

$$\arg \max_{x' s.t. ||x' - x||_\infty \leq \delta} L(\theta, x', y) = x + \arg \max_{||\eta||_\infty \leq \delta} L(\theta, x + \eta, y)$$

given a data point x , a limit δ , and a loss function $L(\theta, x, y)$, to find the 'worst' adversarial example (maximizes loss) in the L_∞ -ball around x **TODO** ▶ *cost vs loss function?? Rewrite without x' , but with ϵ ?*◀

To do that, Goodfellow et al. linearize the loss function around x , giving us

$$\arg \max_{||\eta||_\infty \leq \delta} L(\theta, x + \eta, y) = \arg \max_{||\eta||_\infty \leq \delta} L(\theta, x, y) + \nabla_x L(\theta, x, y)^T \eta = \arg \max_{||\eta||_\infty \leq \delta} \nabla_x L(\theta, x, y)^T \eta$$

We can easily see that $\eta = \delta \cdot \text{sign}(\nabla_x L(\theta, x, y))$ is a valid maximizer of the scalar product $\nabla_x L(\theta, x, y)^T \eta$. This corresponds to a single, as large as possible step in the direction of the gradient.

The method also yielded good results... **TODO** ▶ *"good"??*◀ but worse than L-BGFS (cite DVERSARIAL MACHINE LEARNING AT SCALE)

Since this approach only needs one backpropagation step instead of iterations like L-BGFS algorithm run, this was called the Fast Gradient Sign Method [find source again, was in one of them]. It made adversarial training feasible, which aims to improve a model's robustness by inserting adversarial examples into the training set.

Variants of FGSM **TODO** ▶ *One-step target class meths, cf DVERSARIAL MACHINE LEARNING AT SCALE*◀

Different variants of FGSM were proposed. [kurakin2017physical], later [kurakin2017scale].

The so-called **basic iterative method** applies FGSM multiple times with a small step size [kurakin2017physical]. The adversarial example is given by

$$X_0^{adv} = X, \quad X_{N+1}^{adv} = \text{Clip}_{X, \epsilon}(X_N^{adv} + \alpha \text{sign}(\nabla_X L(X_N^{adv}, y_{true})))$$

where $\text{Clip}_{X, \epsilon}$ implements pixel-wise constriction to the ϵ -neighborhood of X [kurakin2017physical].

To find adversarial examples of a specific class y_{target} instead of maximizing the loss function, FGSM can be modified to maximize that classes probability $p(y_{target}|X)$. This can be achieved with a single step in direction of $\text{sign}(\nabla_X \log(p(y_{target}|X)))$ [kurakin2017physical], analogous to basic FGSM, or using multiple, like in the aforementioned basic iterative method [kurakin2017scale]. A popular choice for target class is the one originally deemed least likely by the target model. [!!!]

PGD (draft! include more predecessors?) Searching adversarial examples by using projected gradient descent (PGD) as proposed by Madry et al. [madry2017towards] is similar to the basic iterative method. Both use first-order information (!!!mention earlier), i.e. the gradient of the loss function, to iteratively maximize the loss on an input in the neighborhood of the real image. Instead of using multiple FGSM steps, though, Madry et al. use projected gradient descent directly.

[They] formulate the adversary's goal as... **TODO** ▶ *include the dual formulation? Not that important for adv. attacks?*◀ Note that the formulation by [FGSM guys] is encompassed in this, by using $S=...$,

then linearizing this maximization problem ... [They] argue that this is insufficient to solve the optimization problem and instead employ PGD to find local maxima in the L_∞ ball. Instead of the single step in gradient direction used in FGSM, we get multiple.

[They] argue that [plateau, maxima], so [it is a good attack]

First-order adversary discussion?

This also ties into adversarial training [dual formula]..? -saddle point formulation (into theory part??) -> adversary goal, L norm flexibility, FGSM as special case

DeepFool DeepFool was introduced by Moosavi-Dezfooli et al. [moosavi2016deepfool] in 2016. The central idea of this iterative is to characterize adversarial vulnerability geometrically: rather than loosely optimizing an attack loss, DeepFool estimates the distance of an input to the closest decision boundary. For affine classifiers, this distance could be calculated in a closed form by projecting the input point onto the decision hyperplane. The highly nonlinear decision boundaries **TODO ▶ hey, is that an assumption they make or where does this come from?◀** of deep neural networks are addressed by DeepFool by exploiting the observation that although the true boundary of a neural network may be highly curved, sufficiently small neighborhoods can often be approximated accurately by a hyperplane; the shortest path to this approximate boundary is assumed to provide a good estimate of the minimal adversarial perturbation. This auxiliary hyperplane is iteratively calculated by their **TODO ▶ who's they?◀** first-order Taylor expansion, shifting closer and closer to the actual nonlinear decision boundary until eventually passing it. In mathematical terms, at an intermediate point x_i , the classifier is iteratively replaced by its local linear approximation,

$$f(x) \approx f(x_i) + \nabla f(x_i)^T (x - x_i), \quad (4)$$

which defines a locally linear decision boundary. The perturbation is obtained by projecting the current point onto this locally estimated boundary, yielding an update

$$x_{i+1} = x_i + r_i, \quad (5)$$

where r_i is the smallest perturbation that crosses the linearized boundary. The procedure is repeated until the classifier changes its prediction.

For multiclass classifiers, DeepFool extends this idea by considering the set of pairwise decision boundaries between the current class and competing classes. At each iteration, the method computes the locally closest boundary under the linear approximation and updates the input toward that boundary.

DeepFool is made several important contributions to the field of adversarial attacks, the first major one being that it provided a practical approximation of minimal adversarial perturbations, making adversarial robustness palpable in terms of distance to the decision boundary. Secondly, it introduced a geometric perspective on adversarial attacks: instead of treating adversarial generation as a generic constrained optimization problem, it directly exploits the local structure of the classifier. Third, this formulation leads to substantially improved efficiency compared with previous optimization-based attacks such as the L-BFGS attack of Szegedy et al. [szegedy2013intriguing], which required solving a computationally expensive optimization problem for each input. By using gradient information to directly estimate the boundary geometry, DeepFool achieves smaller perturbations than earlier fast gradient-based methods while requiring considerably less computation than general-purpose optimization approaches. **TODO ▶ compare with PGD, FGSM, ...◀**

Carlini-Wagner The Carlini-Wagner (C&W) attack is an influential attack presented by Carlini et al. [carlini2017towards]. The method is building directly upon the original L-BFGS method by Szegedy et al. [szegedy2013intriguing] by substantially improving the objective function of the optimization problem. They further introduced a trick to formulate the problem as an unconstrained optimization task, instead of a constrained one.

Just like L-BFGS, given an input example x and a target class t , the attack seeks an adversarial example $x' = x + \delta$ by solving:

$$\min_{\delta} \|\delta\|_p + c \cdot f(x + \delta), \quad (6)$$

where $\|\delta\|_p$ measures the perturbation magnitude and $f(\cdot)$ is an objective function that encourages the model to classify the perturbed input as the target class. The key contribution of the C&W attack is the use of a logit-based loss function rather than optimizing the softmax output probabilities. The loss is defined as:

$$f(x') = \max \left(\max_{i \neq t} Z_i(x') - Z_t(x'), -k \right), \quad (7)$$

where $Z_i(x')$ denotes the logit corresponding to class i , and k is a confidence parameter controlling the margin by which the target class must exceed competing classes. This formulation directly optimizes the decision boundary of the classifier **TODO** ► does it change the classifier? ◀ and avoids the gradient saturation problems associated with softmax probabilities.

To address the pixel-range constraint of image inputs, the C&W attack introduces a clever variable transformation:

$$x' = \frac{1}{2}(\tanh(w) + 1), \quad (8)$$

where w is an unconstrained latent space optimization variable. This transformation converts the original constrained optimization problem into an unconstrained one while guaranteeing that the generated adversarial example remains within the valid input range. The attack then optimizes w using gradient-based optimization methods, typically the Adam optimizer[**kingma2014adam**].

The original work introduced variants based on different distance metrics, including L_0 , L_2 , and L_∞ norms, allowing the attack to evaluate robustness under different threat models.

The effectiveness of the C&W attack also demonstrated weaknesses in several proposed defenses. In particular, it was shown to successfully bypass defense distillation [**papernot2016distillation**], a technique proposed to improve adversarial robustness by training a model using softened output probabilities from a teacher network. Although defense distillation appeared to increase resistance against earlier attacks, Carlini and Wagner demonstrated that the defense distillation method did in fact only obscure the model's gradient albeit not truly hiding it **TODO** ► So what? How is this relevant? elaborate maybe ◀. By directly targetting the logit with their loss function, Carlini et al. were able to efficiently generate adversarial examples in spite of defense distillation by directly approximating the obscured decision boundary.

ZOO (draft!) Zeroth-Order Black Box Attacks (ZOO) [**chen2017zoo**] In 2017, Chen et al. [**chen2017zoo**] proposed Zeroth-Order Black Box Attacks (ZOO), a black-box attack without the use of a substitute model. This way, they avoided losses in transferability to the target and achieved comparable **TODO** ► elaborate ◀ results to then-state of the art white box attacks.

This attack is possible if the adversary is allowed to query the target model, i.e. obtaining the function value, here confidence scores, $f(x)$ for an arbitrary x . This might be the case for a classification-as-a-service tool. Gradients and model configuration, however, are not available (black box).

To circumvent this restriction, Chen et al. estimate a pseudo-gradient using two queries per coordinate:

$$\hat{g}_i := \frac{\partial f(x)}{\partial x_i} \approx \frac{f(x + he_i) - f(x - he_i)}{2h}$$

where f is a loss function defined on the model's output and h is set to a small constant instead of approaching 0 like in the derivation formula. Naively, this would suffice to estimate full gradients and perform gradient descent. However, this formula requires two evaluations per coordinate, $f(x + he_i)$ and $f(x - he_i)$. Since each pixel translates to three coordinates of input data, this is not a feasible approach.

Instead, Chen et al. perform stochastic coordinate descent, improving just a batch of coordinates at each step. This way, they do not need to estimate the full gradient each time, similar to how stochastic gradient descent only uses a batch of samples at each step. For the optimization itself, they use ADAM or Newton's method.

Still, we need to find a suitable loss function f . In fact, we can use different loss functions to adapt different white-box attacks. In their work, Chen et al. adapted the Carlini-Wagner attack. C-W's loss function performs on the raw logits of a model, an information inaccessible in the black-box case.

Instead, Chen et al. use the logarithm of the confidence scores as a proxy of sorts for the logits

TODO ▶ *argue* ◀ and obtain

$$f(x, t) = \max\{\max_{i \neq t}\{\log[F(x)]_i\} - \log[F(x)]_t, -\kappa\}$$

for targeted attacks and

$$f(x, t) = \max\{\log[F(x)]_{t_0} - \max_{i \neq t_0}\{\log[F(x)]_i\}, -\kappa\}$$

for untargeted ones. **TODO** ▶ *mutually improve this with Carlini-Wagner* ◀

Additionally, they propose different measures to speed up optimization even more **TODO** ▶ *...* ◀

TODO ▶ *Newton method using Hessian, and other tricks* ◀

2.4 Notable Poisoning Attacks

Poisoning attacks were already included in [The security of machine learning...], which clustered attacks on machine learning into exploratory and causative, analogous to our evasive and poisoning taxonomy.

The first poisoning attack that got greater attention was Biggio et al.'s gradient-based attack on Support Vector Machines. After, more research was contributed, especially on attacks on neural networks.

Backdoor poisoning attacks With backdoor attacks on machine learning as proposed by Chen et al. [chen2017targeted] and Gu et al. [gu2017badnets], the adversary's goal is to introduce a backdoor for himself into the model. While the performance on normal inputs is to be kept up, the model should react to specific inputs by misclassifying them. (Either generally, or targeted.) These specific inputs carry a special pattern, the "key" or "trigger".

Gu et al. consider an adversary that has full access to the model or a base version of it. Namely, an ML-as-a-service provider (full access) and a provider of pretrained models (base version access). In both cases, the architecture is specified already, as well as a training set. A backdoor key pattern is chosen by the attacker. The adversary has to find model weights performing well on general samples from the set yet vulnerable to the backdoor.

For that, Gu et al. use training set poisoning, adding a fraction of training images to the dataset again, only with the trigger pattern included and the labels changed according to varying strategies. They did not yet derive a way to optimize poison samples, instead, pattern, label change formula and infected fraction need to be selected by the adversary.

Chen et al. instead use a black-box threat model where the adversary does not have access to model architecture or training data and can only inject a small number of adversary samples including arbitrary labels. The key should not be easily noticeable by a human observer. Crucially, the model performance on non-backdoor inputs should be kept up.

Backdoor keys may either be a single instance or a pattern added to instances. The former ("input-instance-key" strategy) allows injecting just very few poisoned examples while the latter ("pattern-key") can be used to craft adversarial examples from a wide range of inputs, which is often more useful to a hypothetical attack.

In input-instance-key strategies, Chen et al. propose adding different noised versions of the key-to-be to the training set, each with the chosen label. The idea is to get the model to overfit to these.

Pattern-key strategies are more complex. The adversary can either **TODO** ▶ *cont.* ◀

Like Gu et al.'s attack, theirs also depends on hyperparameter choices only, with the framework not optimizing poisoning on its own.

2.5 Clean-Label attacks

Notably, while the backdoor poisoning attack described by Chen et al. does not need access to the model, the adversary has to be able assign arbitrary labels to samples, as does the one by Gu et al. However, this can be highly noticeable and alert human overseers to the attack or might just not be feasible in another way. Clean-label attacks do not give such power to the adversary.

One such attack was proposed by Shafahi et al. [shafahi2018poison]. In contrast to Chen et al.'s attack, the adversary does know the model parameters, which might happen if the target model is based on a pretrained, freely accessible one. His goal is to have a target instance misclassified without modifying it, e.g. getting a malware script to bypass a ML antivirus.

First, he chooses a base instance, such that its class, the "base class", is the one he wants the target instance to be classified as. With this, he crafts a poison instance p by solving

$$p = \arg \min_x ||f(x) - f(t)||_2^2 + \beta ||x - b||_2^2$$

where the first term compares the target t and poison candidates x in feature space (the more similar, the better), and the second penalizes difference from the base instance. The general idea is to find a poison instance close to b , therefore likely to be labeled the same, yet close to t in feature space (output of second-to-last layer), so t is treated the same as b by the model. Shafahi et al. argue that a suitable x can be found due to f being supposedly highly nonlinear. For optimization, two steps are iterated: First, a gradient step ("forward step"), then a step towards b ("backwards step").

► FAIL graphic - similar ◀

Clean-label backdoor attacks Saha et al. [saha2020hidden] made an effort to combine clean-label attacks with backdoor poisoning attacks, albeit Turner et al. did the same a year earlier [turner2019label]. Both also considered how to hide the trigger during poisoning to decrease the noticeability of the backdoor, though differently. Turner et al. manually select a "trigger amplitude", essentially an opaqueness. Saha et al., on the other hand, directly incorporate it into their optimization and not even have the pattern itself in the poisoned instance.

Saha et al. consider a threat model in which the adversary can add poison to a dataset the victim uses for fine-tuning a pretrained model, which the adversary has access to. The scenario is similar to the one described by Gu et al., with the difference that the adversary attacks the fine-tuning, not the pretraining.

The attack itself is similar to Shafani et al.'s. The adversary chooses a source and a target image from the data distribution. He then applies his trigger to the source, obtaining the patched source. Like before, the poison sample is optimised to be close to the target in pixel space (so it probably receives the target label) while close to the patched source in feature space (so the model has difficulties distinguishing them). The notation is flipped, though...

Let $\tilde{s}$ be the patched source, t the target, and ϵ a small, chosen threshold. Saha et al. optimize

$$\arg \min_x ||f(x) - f(\tilde{s})||_2^2 \quad \text{s.t.} \quad ||f(x) - f(t)||_\infty^2 < \epsilon$$

to obtain the poisoned sample x . Again, f is the output of the second-to-last layer, the feature space representation. To optimize, they use projected gradient descent (2.3). However, in each optimization step, a new randomly-patched source is selected so the backdoor generalizes to arbitrary images and patch placements. ...

A more recent work by Zeng et al. [zeng2023narcissus] addresses issues with prior work on clean-label backdoor attacks. They consider a black-box threat model where the adversary can only poison a small subset of the training data. The adversary chooses the target class and knows some examples in it. He also has general information on the task the victim model should perform, such as "face recognition", but not specifics like classes used. This enables the attacker to draw up more data instances related to the task, though they might not belong to the original data distribution and are therefore referred to as "public out of distribution (POOD) examples". This is especially easy in image or language tasks where lots of samples are available in the web.

All in all, the adversary is considerably weaker than in aforementioned attacks.

Their approach revolves around optimizing the trigger pattern used under a constraint so it is harder to notice for humans (as in evasion attacks). Suppose the attacker had access to the model the victim would obtain without poisoning. We call this model the oracle f_{orc} . Recall that he also has access to a dataset of the target class D_t . Zeng et al.’s approach is to choose a trigger δ^* maximizing the oracle’s confidence on the target class, i.e. solving

$$\delta^* = \arg \min_{\delta \in \Delta} \sum_{(x,t) \in D_t} L(f_{orc}(x + \delta), t)$$

where Δ is the set of allowed perturbations, e.g. the L_∞ -ball. This way, δ^* itself is related to the target class, and adding it to any input increases the likelihood of that input being classified as the target class.

However, the adversary does not have access to f_{orc} . Instead, Zeng et al. propose training a surrogate model f_{sur} on the POOD data D_{POOD} . The above equation becomes

$$\delta^* = \arg \min_{\delta \in \Delta} \sum_{(x,t) \in D_t} L(f_{sur}(x + \delta), t)$$

Now, the attack works as follows. First, f_{sur} is trained using D_{POOD} and D_t . While training the surrogate model on both D_{POOD} and D_t simultaneously would be possible, Zeng et al. suggest training on D_{POOD} first and fine-tune on D_t to make modification for a possible second attack on another class more efficient.

In the second step, they use (stochastic) projected gradient descent to find the trigger δ^* and subsequently add it to a portion of the target class samples D_t . The victim is then given this poisoned target class data set.

In a backdoor exploit, the adversary can amplify the trigger to

3 Discussion

3.1 Real-World and Physical Adversarial Attacks

One might expect the study of adversarial attacks to be driven primarily by security—the need to defend deployed systems against real attackers. In practice, confirmed *malicious* adversarial-example attacks against real institutions or individuals remain strikingly rare. [biggio2018wild; raff2023dont] Two explanations suggest themselves. The first is that such attacks slip through unnoticed: a successful adversarial input may be logged as an ordinary misclassification, ascribed to generic model error, or filed under an unrelated incident category, so that its true incidence is undercounted. The second, which we find more plausible, is that adversarial examples are simply of little use to an attacker. Crafting them reliably demands specialised machine-learning and security expertise, whereas cruder techniques—distributed denial-of-service, character obfuscation to slip past spam filters, or merely packing a malware binary—remain cheap, fast, and effective. [apruzzese2023real; raff2023dont] Building an adversarial example to defeat a system that a simpler attack would already defeat is like learning to pick a lock when the crowbar is cheaper, quicker, and works just as well; tellingly, the few confirmed real-world cases, such as commercial CAPTCHA-solving services or the prompt injection of deployed chatbots, tend to be those where no such simpler alternative exists. The relevance of adversarial attacks as an object of study is therefore less that of an urgent, widespread security threat than an *epistemic* one: they serve as a lens on how machine perception diverges from human perception.

This does not render the phenomenon academic in the pejorative sense. As machine learning is embedded ever more deeply into the physical world—autonomous vehicles, automated border control, biometric surveillance—the question of whether an input that is semantically unchanged to a human can nonetheless fool a sensor pipeline becomes a central engineering concern. The body of work on *physical* adversarial attacks shows that it can: inconspicuous stickers on a stop sign reliably cause targeted misclassification, both in the laboratory and from a moving vehicle; [eykholt2018robust] printable adversarial patches force a chosen label onto almost any scene they appear in; [brown2017adversarial] adversarial eyeglass frames defeat face

recognition;[[sharif2016accessorize](#)] and spoofed sensor returns or crafted road-surface markings can mislead the perception stack of a production self-driving system.[[cao2019adversarial](#); [tencent2019tesla](#)]

Such results matter because robustness is a core requirement for any safety-relevant deployment—which makes it all the more pressing to understand *why* such systems fail, and what that failure implies. It is to those questions that we now turn.

3.2 Explaining Adversarial Examples

Having seen *where* adversarial examples arise, we now turn to *why* they exist, tracing how successive attempts to explain the phenomenon have replaced one another since its discovery in 2013.

Szegedy et al. (2013): the discovery. The work that introduced adversarial examples did not yet offer a concrete explanation, but it established two observations that every later account had to reckon with.[[szegedy2013intriguing](#)] First, adversarial examples are neither random artefacts nor isolated anomalies: they occupy structured regions of the input space—“pockets” in the data manifold—that are sparse, and therefore hard to encounter through random sampling, yet can be produced reliably through optimisation. Second, the effect is not merely overfitting. The same adversarial input often fools different models with different hyperparameters, and even different architectures—the phenomenon later called *transferability*—which already argued against attributing adversarial examples to a single model overfitting its training set.

Goodfellow et al. (2014): the linear explanation. The account of Goodfellow et al. [[goodfellow2014explaining](#)] became one of the most influential in the field. It overturns the intuition, inherited from [[szegedy2013intriguing](#)], that adversarial examples stem from the *extreme non-linearity* of deep networks—highly contorted decision surfaces riddled with rare, low-probability blind spots—possibly compounded by overfitting. Goodfellow et al. argue the opposite: linear behaviour in high-dimensional spaces is already *sufficient* to produce adversarial examples. Because modern networks (ReLU, maxout, LSTM) are deliberately built to behave in near-linear ways for ease of optimisation, the argument carries over from strictly linear models to deep networks.

The mechanism is a high-dimensional “bloating” effect. Feature precision is limited—an 8-bit pixel discards everything below $1/255$ of the dynamic range—so a classifier ought to be insensitive to any perturbation η with $\|\eta\|_\infty < \epsilon$. For a linear model, however, $w^\top(x + \eta) = w^\top x + w^\top \eta$, so the perturbation shifts the activation by $w^\top \eta$. Maximising this shift under the max-norm budget $\|\eta\|_\infty \leq \epsilon$ yields $\eta = \epsilon \operatorname{sign}(w)$, so that with n input dimensions and mean weight magnitude m ,

$$w^\top \eta = \epsilon \sum_{i=1}^n |w_i| = \epsilon m n.$$

The per-coordinate change ϵ remains fixed and imperceptible, yet the induced change in the activation grows *linearly* with the dimension n . In sufficiently high dimensions, many individually imperceptible, weight-aligned changes accumulate into one large change in the output. Figure 4 illustrates this: the per-pixel budget stays flat as n grows, while the change in the model’s output climbs until it crosses the threshold that flips the prediction.

This point is easily overstated, and often is: Goodfellow et al. do *not* claim that networks *are* linear, nor that linearity is a *necessary* ingredient of the explanation. They show only that non-linearity is not *necessary*—the simplest possible model, plain logistic or softmax regression with no hidden layer, already exhibits adversarial examples—so that, by a parsimony argument, the more elaborate non-linearity hypothesis can be dispensed with. (The refined, later view holds that both dimensionality and local curvature matter, as reflected in the stronger iterative and minimum-norm attacks discussed above.[[madry2017towards](#); [moosavi2016deepfool](#)])

The linear view also provided the first explanation of transferability. Since adversarial examples now occupy broad, contiguous half-spaces rather than fine pockets, different models’ adversarial regions overlap heavily, which explains why an example crafted for one model tends to fool another at all. And because the discriminative direction is largely a property of the data rather than of the model, models trained on the same task tend to learn similar weight vectors, so that the same perturbation pushes them toward the same incorrect class—which explains why the models tend to

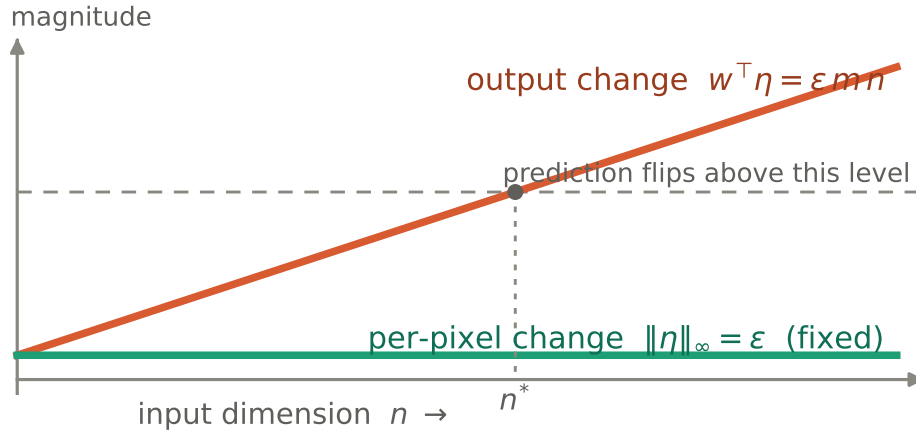


Figure 4: The linear explanation of adversarial examples. Under an L_∞ (per-pixel) budget every input coordinate is perturbed by at most ϵ , so the perceptual size of the perturbation, $\|\eta\|_\infty = \epsilon$, stays fixed as the input dimension n grows. The induced change in the model’s activation, $w^\top \eta = \epsilon m n$, instead grows *linearly* with n . Once the dimension is large enough (at n^*) this accumulated change crosses the threshold that flips the prediction—so in high dimensions an imperceptible perturbation suffices.

512 agree on that class. As a demonstration, Goodfellow et al. proposed the Fast Gradient Sign Method
 513 (Section 2.3); beyond its practical value, the fact that so cheap a first-order—that is, *linear*—step
 514 reliably fools many different models is itself evidence for the linearity hypothesis.

515 Ilyas et al. (2019): *features, not bugs*. Ilyas et al. [ilyas2019adversarial] shift the framing once more,
 516 recasting adversarial examples not as pathological failures but as a *mismatch between machine and*
 517 *human perception*: models exploit patterns that are genuinely predictive, but imperceptible to, and
 518 unintuitive for, humans. They distinguish *predictive* features (features correlated with the label)
 519 from *robust* features (features that remain predictive under perturbation). Since empirical risk
 520 minimisation rewards any signal that improves accuracy, a learner will readily rely on predictive
 521 but non-robust features; adversarial vulnerability is then an expected consequence of standard
 522 training, rooted in the gap between the human sense of similarity and the statistical structure of the
 523 data. Their striking demonstration makes this concrete: they relabel every image of CIFAR-10 with
 524 a new, random target class and perturb it so that a model reads the new label, while to a human
 525 the images appear unchanged and hence entirely mislabelled. A standard model trained on this
 526 dataset nevertheless generalises well to the *clean*, unmodified test set—evidence that it has latched
 527 onto real, transferable features that merely happen to be imperceptible to us. This view also bears
 528 on interpretability: if a model succeeds by relying on human-imperceptible features, there may be
 529 little in its decisions for a human to understand.

530 3.3 Pertinence of Adversarial Examples as a Subject Area

531 The two preceding subsections approach adversarial examples from opposite ends—one asking
 532 whether they pose a real danger, the other asking why they exist—yet they converge on the same
 533 conclusion: the primary significance of adversarial examples is *epistemic*. As a live security threat
 534 they have so far been marginal, dwarfed by cruder and cheaper forms of attack; [apruzzese2023real;
 535 raff2023dont] their importance lies less in what they let an attacker do than in what they reveal
 536 about the systems they target.

537 What they reveal is considerable. Before the discovery and systematic study of adversarial
 538 examples, machine-learning models were, as a rule, evaluated almost exclusively on accuracy, and
 539 robustness was expected only implicitly. [biggio2018wild; hendrycks2019benchmarking] Adver-
 540 sarial examples showed that high test accuracy does not imply that a model has grasped the concept
 541 a human would: a classifier will confidently mislabel an input that is, to a human observer, seman-
 542 tically unchanged. This forced a more intricate account of how machine learners operate—from

Goodfellow et al.’s high-dimensional linearity to Ilyas et al.’s reframing of adversarial features as genuine but human-imperceptible signal[goodfellow2014explaining; ilyas2019adversarial]—and, with it, a redefinition of *robustness* from an assumed by-product of accuracy into a distinct property that must be explicitly demanded, engineered, and evaluated. This shift is not merely academic: as machine learning is embedded in autonomous vehicles, border control, and surveillance, physical adversarial attacks make plain that robustness cannot be taken for granted in exactly those deployments where its absence would be most costly.

Viewed from a distance, the lesson generalises well beyond the field. When we trust a simple mathematical proxy to stand in for a messy real-world objective—as an overriding focus on accuracy does for machine learners—we are liable to be disappointed, and the resulting mismatch can surface suddenly and in unexpected ways. The enduring contribution of adversarial examples is to have made that mismatch—between human and machine notions of similarity and understanding—impossible to ignore.

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